Literature-Implied Status Note for arXiv:1903.01761

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Status. `literature_implied_answer` (partial subcases/status update). This is not an original proof and not a full answer to the unrestricted Daugavet tensor-product question.

Source/open-problem metadata. Abraham Rueda Zoca, Pedro Tradacete, and Ignacio Villanueva, "Daugavet property in tensor product spaces," arXiv:1903.01761. In the introduction, Question 1 asks whether, whenever Banach spaces X and Y both have the Daugavet property, the tensor products

$$X\widehat{\otimes}_\varepsilon Y \quad \text{and} \quad X\widehat{\otimes}_\pi Y$$

also have the Daugavet property.

Supporting projective WODP metadata. Miguel Martin and Abraham Rueda Zoca, "Daugavet property in projective symmetric tensor products of Banach spaces," arXiv:2010.15936; Banach J. Math. Anal. 16 (2022), article 35. Section 5 introduces the weak operator Daugavet property (WODP). Theorem 5.4, labelled `theorem:WODP` in the source, proves that if X and Y have WODP, then

$$X\widehat{\otimes}_\pi Y$$

has WODP. Since WODP implies the Daugavet property, this gives a positive projective-tensor subcase of the source question under the stronger WODP hypothesis.

Supporting injective WODP metadata. Abraham Rueda Zoca, "Weak operator Daugavet property and weakly open sets in tensor product spaces," arXiv:2407.21585. Theorem `theo:daugainjective` proves that if X^* and Y^* have WODP, then

$$X\widehat{\otimes}_\varepsilon Y$$

has the Daugavet property. Remark `remark:mejoratrada` explicitly says that this improves the $L_1(\mu)\widehat{\otimes}_\varepsilon L_1(\nu)$ case proved as Theorem 1.1 in arXiv:1903.01761.

Supporting transfinite projective metadata. Antonio Aviles, Johann Langemets, Miguel Martin, and Abraham Rueda Zoca, "Transfinite Daugavet property," arXiv:2604.14102. In the section "Tensor products," Theorem `theo:tensorprolargeDP` proves that for an infinite cardinal κ, if X and Y are injective Banach spaces with the κ-Daugavet property, then

$$X\widehat{\otimes}_\pi Y$$

has the κ-Daugavet property; the same is proved for the κ-perfect Daugavet property.

Identification. The 2022 WODP theorem directly answers the projective half of the source question for WODP factors. The 2024 WODP theorem directly answers the injective half when both dual spaces have WODP, and the supporting paper explicitly identifies this as an improvement of the 1903 L_1-space theorem. The 2026 theorem gives a further projective-tensor subcase for injective spaces with a transfinite strengthening of the Daugavet property. These are genuine positive pieces around Question 1, but they do not remove the stronger hypotheses.

Scope limitations. The unrestricted problem remains open in this packet: no checked supporting source proves that $X\widehat{\otimes}_\pi Y$ or $X\widehat{\otimes}_\varepsilon Y$ has the Daugavet property for arbitrary Daugavet spaces X, Y. This packet also does not settle whether every Daugavet space has WODP or ODP.

Search evidence. The run indexes were searched for `1903.01761`, `Daugavet property`, `projective tensor`, `injective tensor`, `WODP`, and `weak operator Daugavet`. No existing packet covered this target. Local source search then identified arXiv:2010.15936, arXiv:2407.21585, and arXiv:2604.14102 as the decisive later literature for these scoped positive answers.

References

[1] A. Rueda Zoca, P. Tradacete, and I. Villanueva, "Daugavet property in tensor product spaces," J. Inst. Math. Jussieu 20 (2021), 1409–1428; arXiv:1903.01761.

[2] M. Martin and A. Rueda Zoca, "Daugavet property in projective symmetric tensor products of Banach spaces," Banach J. Math. Anal. 16 (2022), article 35; arXiv:2010.15936.

[3] A. Rueda Zoca, "Weak operator Daugavet property and weakly open sets in tensor product spaces," arXiv:2407.21585.

[4] A. Aviles, J. Langemets, M. Martin, and A. Rueda Zoca, "Transfinite Daugavet property," arXiv:2604.14102.